

一、DAY1

1. 计算: $1\frac{5}{7} \times (5 \div \frac{5}{6} - \frac{5}{6} \div 5) = \underline{10}$.

$$\begin{aligned}\text{原式} &= 1\frac{12}{7} \times (5 \times \frac{6}{5} - \frac{5}{6} \times \frac{1}{5}) \\ &= 1\frac{12}{7} \times (6 - \frac{1}{6}) \\ &= 1\frac{12}{7} \times \frac{35}{6} \\ &= \frac{23}{7} \times \frac{35}{6} \\ &= 10\end{aligned}$$

2. 计算: $2\frac{1}{3} \times \frac{2}{7} - (\frac{5}{7} - \frac{1}{3}) = (\underline{B})$.

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{3}{7}$

D. $\frac{4}{7}$

$$\begin{aligned}\text{原式} &= \frac{7}{3} \times \frac{2}{7} - (\frac{5}{7} - \frac{1}{3}) \\ &= \frac{14}{21} - \frac{8}{21} \\ &= \frac{6}{21} \\ &= \frac{2}{7}\end{aligned}$$

(S 拓展) 3. 计算: $\left[\left(5 - 3\frac{1}{2} \right) \times 2\frac{1}{4} \right] \div \left[4 \div \left(4 + 1\frac{1}{3} \right) \right] = \underline{\hspace{2cm}}$.

$$\begin{aligned}\text{原式} &= [1\frac{1}{2} \times 2\frac{1}{4}] \div [4 \div 5\frac{1}{3}] \\ &= [\frac{3}{2} \times \frac{9}{4}] \div [4 \div \frac{16}{3}] \\ &= \frac{27}{8} \div [\frac{4}{1} \times \frac{3}{16}] \\ &= \frac{27}{8} \div \frac{3}{4} \\ &= \frac{27}{8} \times \frac{4}{3} \\ &= \frac{9}{2}\end{aligned}$$





二、DAY2

1. 填表，分小互化

分数	$\frac{1}{4}$	$\frac{3}{5}$	$\frac{9}{8}$	$\frac{13}{20}$	$2\frac{2}{5}$
小数	0.25	0.6	1.125	0.65	2.4

$$\frac{1}{4} = \frac{1 \times 25}{4 \times 25} = \frac{25}{100} = 0.25$$

$$0.6 = \frac{6}{10} = \frac{3}{5}$$

$$\frac{9}{8} = 1\frac{1}{8} = 1.125$$

$$0.65 = \frac{65}{100} = \frac{13}{20}$$

$$2.4 = 2\frac{4}{10} = 2\frac{2}{5}$$

2. 计算： $2.5 \div 8 + 9.5 \times \frac{1}{8} + 4 \times 0.125 = (A)$.

A. 2

B. 3

C. 4

D. 5

$$\text{原式} = \frac{5}{2} \times \frac{1}{8} + \frac{19}{2} \times \frac{1}{8} + \frac{1}{4} \times \frac{1}{8}$$

$$= \frac{5}{16} + \frac{19}{16} + \frac{1}{16}$$

$$= \frac{24}{16} + \frac{1}{16}$$

$$= \frac{25}{16} + \frac{1}{16}$$

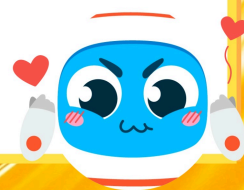
$$= 2$$

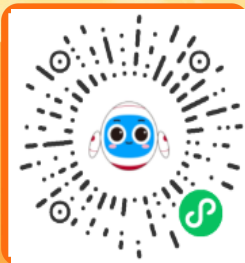
(S 拓展) $3.359 \times \frac{357}{358} = \frac{357}{358} \overline{) 357}$

$$\text{原式} = (358 + 1) \times \frac{357}{358}$$

$$= 358 \times \frac{357}{358} + 1 \times \frac{357}{358}$$

$$= \frac{357}{358} \overline{) 357}$$





三、DAY3

1. 下列小数化成分数不正确的是 (B) .

✓ A. $0.12 = \frac{3}{25}$

B. $0.1 = \frac{1}{9}$ ✗

✓ C. $2.5 = \frac{5}{2}$

$0.12 = \frac{12}{100} = \frac{3}{25}$

$0.1 = \frac{1}{10}$

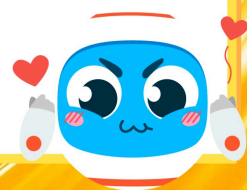
$2.5 = \frac{25}{10} = \frac{5}{2}$

2. 计算: $0.4 \times \left[\frac{11}{52} \div 2\frac{3}{4} \times (4.3 - 1.8) \right] \times 26 = \underline{2}$.

$$\begin{aligned} \text{原式} &= 0.4 \times \left[\frac{11}{52} \div \frac{11}{4} \times 2.5 \right] \times 26 \\ &= 0.4 \times \left[\frac{11}{52} \times \frac{4}{11} \times 2.5 \right] \times 26 \\ &= 0.4 \times \left[\frac{2}{26} \times 2.5 \right] \times 26 \\ &= (0.4 \times 2.5) \times \left(\frac{2}{26} \times 26 \right) \\ &= 1 \times 2 \\ &= 2 \end{aligned}$$

(S 拓展) 3. $2006 \times 2008 \times \left(\frac{1}{2006 \times 2007} + \frac{1}{2007 \times 2008} \right) = \underline{2}$.

$$\begin{aligned} \text{原式} &= 2006 \times 2008 \times \frac{1}{2006 \times 2007} + 2006 \times 2008 \times \frac{1}{2007 \times 2008} \\ &= \frac{2008}{2007} + \frac{2006}{2007} \\ &= \frac{2008 + 2006}{2007} \\ &= \frac{2 \times 2007}{2007} \\ &= 2 \end{aligned}$$





四、DAY4

$$1. \frac{7}{6 + \frac{5}{4 + \frac{3}{2}}} = \frac{77}{76}$$

$$\begin{aligned} 5 \div (4 + \frac{3}{2}) &= 5 \div (\frac{8}{2} + \frac{3}{2}) = 5 \div \frac{11}{2} = 5 \times \frac{2}{11} = \frac{10}{11} \\ 7 \div (6 + \frac{10}{11}) &= 7 \div 6\frac{10}{11} = 7 \div \frac{76}{11} = 7 \times \frac{11}{76} = \frac{77}{76} \end{aligned}$$

$$2. \text{计算: } \left[\frac{4}{5} + \frac{3}{4} \times \left(\frac{5}{12} - 0.15 \right) \right] \div 1\frac{1}{8} = \frac{40}{27}$$

$$\begin{aligned} \text{分子: } & \left[\frac{4}{5} + \frac{3}{4} \times \left(\frac{5}{12} - \frac{3}{20} \right) \right] \div \frac{9}{8} \\ &= \left[\frac{4}{5} + \frac{3}{4} \times \left(\frac{25}{60} - \frac{9}{60} \right) \right] \times \frac{8}{9} \\ &= \left[\frac{4}{5} + \frac{3}{4} \times \frac{16}{60} \right] \times \frac{8}{9} \\ &= 1 \times \frac{8}{9} = \frac{8}{9} \end{aligned}$$

$$\begin{aligned} \text{分母: } & 4\frac{1}{5} \div 7 = \frac{21}{5} \times \frac{1}{7} = \frac{3}{5} \\ \text{原式: } & \frac{8}{9} \div \frac{3}{5} = \frac{8}{9} \times \frac{5}{3} = \frac{40}{27} \end{aligned}$$

$$(S \text{ 拓展}) 3. \text{计算: } 2018 \div 2019 \frac{1}{2017} = \frac{2017}{2018}$$

$$\begin{aligned} \text{原式} &= 2018 \div \frac{2017 \times 2017 + 1}{2017} \\ &= 2018 \div \frac{(2018+1) \times (2018-1) + 1}{2017} \\ &= 2018 \div \frac{2018 \times 2018 - 1 + 1}{2017} \\ &= 2018 \div \frac{2018 \times 2018}{2017} \\ &= 2018 \times \frac{2017}{2018 \times 2018} \\ &= \frac{2017}{2018} \end{aligned}$$

